📡 Network Devices (10.20.2026)

🌐 What is a Network?

- **Computer Network:** A digital telecommunications network that allows **nodes** to share resources.
- **LAN (Local Area Network):** A network confined to a small geographic area (like an office or home).

🖥️ Endpoints (End Hosts)

End hosts (or **endpoints**) are devices at the edge of the network. They can be **Clients** or **Servers**.

- **Client:** A device that *accesses* a service made available by a server (e.g., your laptop browsing a website).
- **Server:** A device that *provides* functions or services for clients (e.g., a web server hosting a website).
- 💡 **Key CCNA Concept:** A device's role is not permanent; it depends on the transaction.
  - *Example:* If your PC sends a file to a printer, your PC is the client, and the printer is the server. If someone downloads a file from your PC, your PC is acting as the server.

🔀 Network Infrastructure (Switches & Routers)

These devices connect hosts and networks together.

Switch

- Provides connectivity to hosts **within the same LAN**.
- Has **many** network interfaces (ports) for end hosts to connect to.
- *Layer:* Operates at Layer 2 (Data Link).

Router

- Connects **different** networks together (e.g., connecting your LAN to the Internet).
- Has **fewer** network interfaces than a switch.
- *Layer:* Operates at Layer 3 (Network).

🛡️ Network Security (Firewalls)

A **Firewall** monitors and controls network traffic based on configured rules.

- **Host-based Firewall:** Software applications installed *on a specific machine* (like Windows Defender) that filter traffic entering and exiting that host.
- **Network Firewall:** A hardware device placed at the edge of a network to protect the whole network.
- **Next-Generation Firewall (NGFW):** A firewall that combines traditional filtering with more advanced functionalities (e.g., deep packet inspection, intrusion prevention, application awareness).

🏢 Cisco Hardware Cheat Sheet

- **Cisco ASA:** Firewall
- **Cisco Firepower:** Firewall (Next-Gen)
- **Cisco ISR:** Router (Integrated Services Router)

🧪 Practice Quiz & Answers

1. Your company wants to purchase network hardware to which they can plug the 30 PCs in your department. Which is appropriate?

- a) A router
- b) A firewall
- c) A switch
- d) A server

> **Answer: c) A switch** (Switches are for connecting many end hosts within the same LAN).

2. You received a video file from your friend's Apple iPhone using AirDrop. What was his iPhone functioning as?

- a) A server

- b) A client
- c) A local area network

> **Answer: a) A server** (His phone was *providing* the file to you, making it the server in that specific transaction).

3. What is your computer or smartphone functioning as while you watch a YouTube video?

- a) A server
- b) An end host
- c) A client

> **Answer: c) A client** (You are *accessing* a service from YouTube's servers). *Note: It is also technically an "end host," but "client" is the most specific answer for this role.*

4. Your company wants to purchase hardware to connect its separate networks together. What kind of device is appropriate?

- a) A firewall
- b) A host
- c) A LAN
- d) A router

> **Answer: d) A router** (Routers route traffic *between* different networks).

5. Your company wants to upgrade its old network firewall to one that provides more advanced functions. What kind of firewall should they purchase?

- a) A host-based firewall
- b) A next-level firewall
- c) A next-generation firewall
- d) A top-layer firewall

> **Answer: c) A next-generation firewall** (NGFWs include advanced features beyond basic traffic filtering).